

Mirthika G

mirthika2005@gmail.com | 91502 81501 | linkedin.com/in/mirthikag | github.com/mirthikagiri

Education

Chennai Institute Of Technology, B.E Computer Science and Engineering
GPA: 9.47/10

2023 - 2027

Technical Skills

Python, Java, C | HTML, CSS, JavaScript | React, Node.js, Express.js, FastAPI | MongoDB, MySQL PostgreSQL | Numpy, Pandas, Matplotlib, Seaborn, Streamlit | NLP, AI and Machine Learning | Git, Figma | AWS (Cloud Practitioner), Docker.

Projects

- **LegacyMind - Enterprise Knowledge Continuity & Risk Intelligence Platform:** Developed a cloud-native platform that analyzes organizational knowledge distribution and identifies critical employee dependencies to mitigate knowledge-loss risks. Designed a **centralized knowledge management system** using employee, project, and document relationships to model expertise ownership across teams. Implemented **Bus Factor analysis**, documentation coverage tracking, and expertise scoring mechanisms to detect vulnerable teams, knowledge silos, and operational risks. Built scalable document ingestion, search, and risk monitoring workflows using **React, FastAPI, PostgreSQL, AWS S3, and Docker**, enabling centralized knowledge discovery and real-time organizational insights through interactive dashboards.
- **FinSight-A Smart Expense Tracker:** Built a responsive **MERN-based web application** to help users track income, expenses, and category-wise spending patterns in real time. Designed an interactive dashboard that visualizes total income, categorized expenses, and balance trends with CRUD functionality. Integrated with a **Gemini AI chatbot** for **personalized financial guidance**. Used NLP-based intent classification to enable context-aware responses. The chatbot's effectiveness was evaluated using relevance metrics, achieving approximately **85% user relevance** across financial queries. Empowered users to **track and optimize their expenses** while improving overall financial literacy and budgeting awareness.
- **CuraPredict - ICU Bed Demand Prediction:** Developed a data-driven **prediction system** using Python, NumPy, pandas, and scikit-learn. Implemented **Linear Regression, Decision Tree, and Random Forest models** for forecasting. The final model achieved an **R² score of 0.87**. Synthetic data generation and real-world simulation were used to test model robustness under peak-load conditions. The system predicts potential bed shortages in advance, helping **reduce patient wait times** and **improve critical care preparedness**. Focused on improving resource planning through data analysis.

Achievements

JP Morgan&Chase Code for Good Hackathon - **Finalist** (Top 500 out of 50,000+ participants)

Top 5 Finalist, Iteryx Hackathon - Developed AuraI, an AI-powered mental health monitoring and risk assessment platform.

Qualified GATE 2026 (CS & IT) - Validating expertise in core CS subjects and aptitude

Certifications

- AWS Certified Cloud Practitioner (CLF C02)
- Ethical Hacking – NPTEL (Elite Silver)
- Python with Data Structures - NPTEL
- CCNA Networking Essentials/ Programming Essentials in Python - Cisco NetAcad
- Introduction to Modern AI - Cisco NetAcad
- Data Structures and Algorithms Toolbox - Coursera
- Microsoft Azure AI Fundamentals: NLP ,Computer Vision - Microsoft Learn
- MongoDB for Students - MongoDB

Competitive Programming

LEETCODE Solved 435 Problems, Achieved Highest Rating 1884 and secured Top 5.65% Globally.

SKILLRACK Solved 400 problems on SkillRack, enhancing problem-solving and coding efficiency

Extra Curricular

- **Leadership & Event Management:** Served as an **Office Bearer for the Immerse Club** and **Team Lead** of the DGROOVES Dance Crew, leading a 30-member team and managing event logistics, coordination, promotions, choreography, and performance execution. Organized and coordinated multiple college club events, managing event planning, team collaboration, and participant engagement.
- **Attended Grace Hopper Celebration India (GHCI) 2025 :** Participated in technical sessions, networking opportunities, leadership discussions, and interactions with industry professionals.
- **Event Organizer, National-Level CSE Symposium :** Coordinated the DUO Developers competition, ensuring seamless event execution, participant engagement, and operational management.